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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	No debug kaw kaw
Project / Solution Name	JobBud
Name (Leader)	Po Rou Jing
Email & Phone (Leader)	roujing2005@gmail.com +6016-294 3869
University Affiliation(s)	Universiti Malaya
Live Prototype URL	JobBud.app
Code Repository URL (public / access granted to judges)	https://github.com/ngyy111000-beep/JobBud

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

JobBud is a web-based career recommendation system developed using Python and Streamlit and deployed on Replit. It adopts a modular, single-application architecture consisting of five main components, as illustrated in Figure 2.1.

1. User Interface: Developed using Streamlit (app.py), this component provides multilingual support, career assessment questionnaires, CV uploads, and personalised recommendation results.

2. Application Logic: The application processes user information, manages assessment workflows, and coordinates communication between the system modules.

3. Recommendation and Analysis Engines: The rule-based career_engine.py evaluates career suitability using essential skills (35%), transferable skills (25%), knowledge (25%), and software proficiency (15%). Supporting modules perform job-market analysis, skill-gap identification, learning pathway generation, AI-disruption scenario analysis, and SDG contribution mapping.

4. Data Sources and External Services: O*NET datasets provide occupational information, while Himalayas, Remotive, GDELT, and DOSM supply job-market, technology-trend, and salary data. An optional RunPod integration generates explanations without influencing career scores.

5. Data Management and Deployment: Streamlit session state temporarily stores user profiles and assessment results, while local files support data caching. Replit hosts the application, which currently operates without a persistent user database.

Together, these components support integrated career assessment and personalised career-development recommendations.

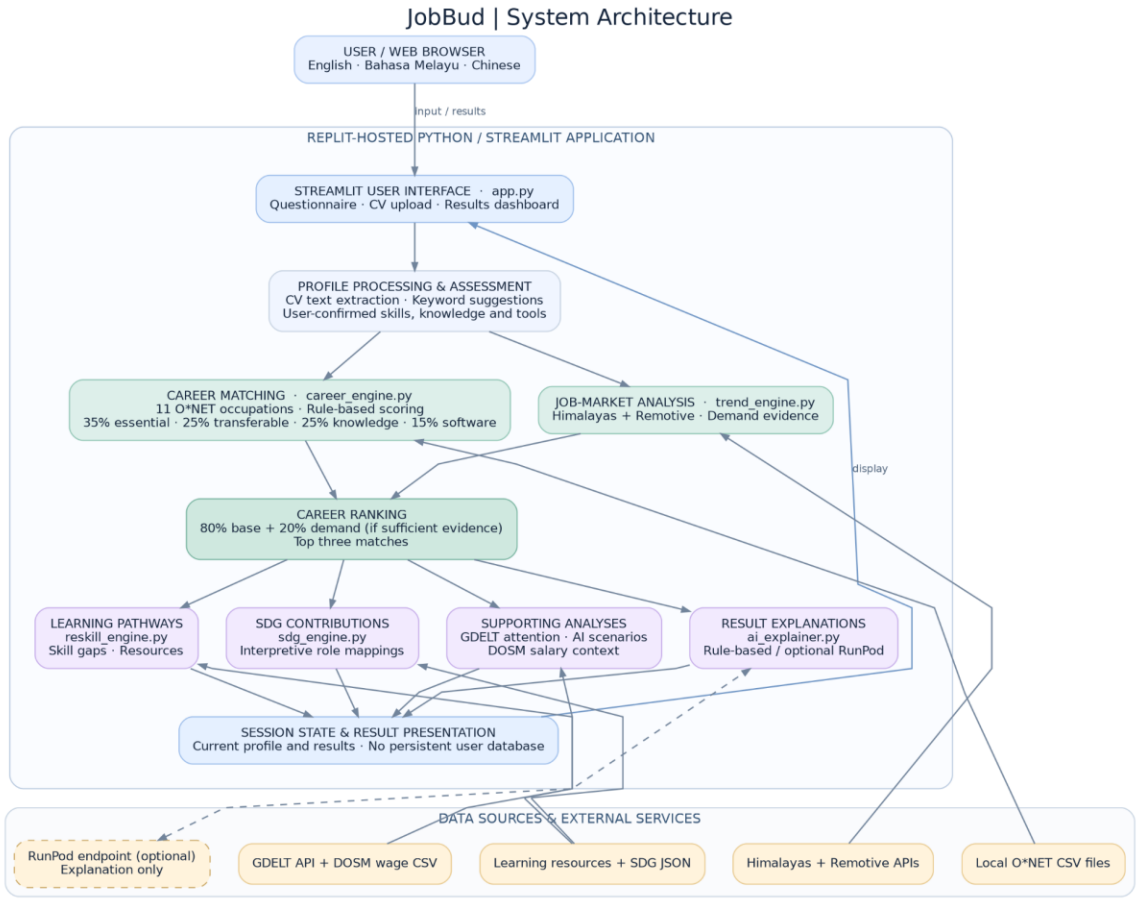


Figure 2.1: JobBud System Architecture Diagram

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python >=3.11	Main programming language for application logic, career matching and data processing	Python Software Foundation License (PSF)
Streamlit >= 1.66.0	Web interface, questionnaires, CV upload and result visualisation	Apache License 2.0

Name & Version	Purpose in Project	License
Pandas >= 3.0.6	Reads and processes O*NET occupational datasets and other CSV files.	BSD 3-Clause
pypdf >= 6.21.2	Extracts text from uploaded PDF CVs	BSD 3-Clause
python-docx >= 1.2.0	Extracts text from uploaded DOCX resumes.	MIT License
requests >= 2.34.2	Makes HTTP requests to external APIs, including the optional RunPod integration	Apache License 2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Remotive Public Jobs API — Remotive	Retrieves remote job listings to analyse labour-market demand and support career recommendations.	Free (Public API)
Himalayas Remote Jobs API — Himalayas	Provides additional remote job listings for job-market analysis and career recommendation adjustments.	Free (Public API)
DOSM Wage Data — Department of Statistics Malaysia	Provides Malaysian industry salary benchmarks for recommended career paths.	Free (Open Access)
GDELT Document API — GDELT Project	Retrieves AI- and technology-related news data to support career disruption scenario analysis.	Free (Open Access)

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale Deployment — Replit	Hosts and deploys the Python-based Streamlit application, enabling public browser access and automatic scaling.	Autoscale Deployment / Region to verify

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None currently active	N/A	N/A	N/A

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
occupation_data.csv	O*NET Database https://www.onetonline.org/	Public	CC BY 4.0	N
essential_skills.csv	O*NET Database	Public	CC BY 4.0	N
transferable_skills.csv	O*NET Database	Public	CC BY 4.0	N
knowledge.csv	O*NET Database	Public	CC BY 4.0	N
software_skills.csv	O*NET Database	Public	CC BY 4.0	N
Remotive Job Listings	Remotive Public API	Public	Provider API terms; attribution required	N
Himalayas Job Listings	Himalayas Jobs API	Public	Provider API terms; attribution required	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
GDELT News Data	GDELT Project	Public	Open use; attribution required	N
Malaysian Industry Salary Data (wages_industry.csv)	OpenDOSM	Public	CC BY 4.0	N
learning_resources.json	Locally curated project file	Self-collected	Team-curated; underlying sources retain their terms	N
sdg_mapping.json	Team-created mappings based on UN SDGs	Self-collected	Team-curated; UN source terms apply	N
User Career Profiles and Skill Assessments	JobBud questionnaire	User-provided	User-provided data; subject to applicable privacy requirements	Y
Uploaded CVs	User uploads (PDF, DOCX, TXT)	User-provided	User-provided data; subject to applicable privacy requirements	Y

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

[/] Yes [ ] No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT (OpenAI; model version varies)	Assisted with code generation, debugging, troubleshooting, improving functionality, technical explanations, and project documentation.	Moderate
Replit Agent	Assisted with application development, code modification, debugging, system architecture analysis, and technical documentation.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team utilised AI-assisted tools to support coding, debugging, and documentation throughout the development of JobBud. AI-generated code and suggestions were reviewed and modified to align with our project requirements. We tested the application's functionalities, identified errors, and refined the implementation based on testing outcomes. Technical documentation was also cross-checked against the actual system architecture and source code. While AI tools improved development efficiency, our team remained responsible for evaluating, integrating, and validating the final code and project deliverables.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

[/] Yes [ ] No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Sponsor-provided credits	Used to develop, test, debug, and deploy JobBud, including its Python-based Streamlit application and career recommendation features.
SDC Organisers / Mentoring Partner	Mentoring sessions	Provided guidance and feedback on project development, innovation, feasibility, and presentation.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit played an important role in developing and deploying JobBud by providing a cloud-based environment for coding, testing, debugging, and hosting the application. It enabled our team to collaborate efficiently and make the prototype accessible through a public website. However, Replit is not essential to JobBud's core functionality, as the application is developed using Python and Streamlit and could be deployed on alternative hosting platforms. The sponsor-provided resources primarily supported development efficiency, accessibility, and deployment rather than the underlying career recommendation algorithms.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

JobBud collects user-provided information through a career assessment questionnaire or optional CV upload, including educational background, skills, knowledge, work experience, and career interests. This information is used solely to analyse users' profiles and generate personalised career recommendations, skill-gap analyses, and learning pathways.

User information and uploaded CVs are processed temporarily during the active session and are not intentionally saved to a database or retained as permanent records. Streamlit session state temporarily maintains assessment information to display the results. JobBud does not currently require user registration or maintain personal assessment histories.

The application uses external APIs to retrieve labour-market information without intentionally sharing users' personal data. The optional RunPod AI integration is not currently active in the inspected implementation. Therefore, no user information is intentionally transmitted to an AI model. Hosting-provider log retention has not been independently verified.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

JobBud has several prototype-stage limitations. Career matching currently covers a limited set of occupations and depends partly on users' self-assessed skills, which may introduce reporting bias. O*NET provides a strong occupational reference framework but primarily reflects the United States labour market and should not be interpreted as direct evidence of Malaysian hiring demand.

Current labour-market signals are derived mainly from remote job-posting APIs and therefore do not represent the complete Malaysian employment market. External APIs may also be affected by rate limits, service availability, or stale data. AI-disruption scenarios use transparent team-defined assumptions rather than predictive machine-learning forecasts.

To mitigate these limitations, JobBud uses explainable weighted scoring, evidence-coverage indicators, source and retrieval information, fallback mechanisms, and explicit limitations. Recommendations are presented as decision-support guidance rather than guaranteed employment outcomes.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [/] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [/] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [/] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [/] Any use of official SDC sponsor resources has been declared in Section 5.
- [/] The prototype described here is accessible to judges through the links provided in Section 1.
- [/] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Signature] PO ROU JING

[Date] 24/10/2026